An outline of the Classification of Finite-dimensional Lie Groups

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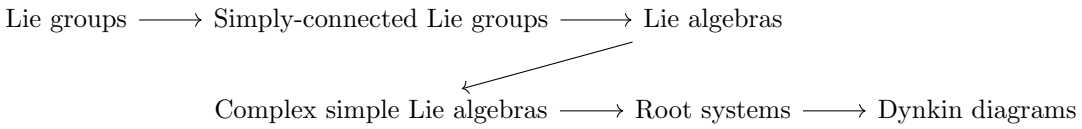
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1 Introduction

Whenever we study some class of objects X, a natural (and illuminating) question to ask is "What are all the X?" For instance, what are all the finite-dimensional vector fields over some base field $\mathbb{K}$? In this case, we know that for every natural number n, we have $\mathbb{K}^n$ and that any other vector space of the same dimension is isomorphic to $\mathbb{K}^n$. The problem of classifying groups (like finite groups, or Lie groups) is more complicated and proceeds in two steps. First, we come up with some suitable definition of "irreducible" with respect to some reduction process, and classify all the irreducible pieces. This is known as the "classification problem." Then, we classify all the ways to make a bigger object given irreducible pieces. This is the "extension problem." For example, the criterion for irreducibility in finite groups is that of being a simple group, the classification problem is the classification of finite simple groups, and the extension problem is to construct a group G given H and N such that $G/N \cong H$. In this paper we will go over the analogous constructions for finite-dimensional Lie groups, and outline the solution of the classification problem (with some steps spelled out more than others).

The basic idea is the following chain of bijections:



To each Lie group we can associate a Lie algebra, and to each Lie algebra we can associate a simply-connected Lie group. Thus to a large extent, studying Lie groups boils down to studying Lie algebras. Lie algebras can further be decomposed into "semisimple" components, which can then be decomposed into "root spaces." Root spaces are combinatorially restricted, and we can enumerate all possible irreducible root spaces. From then, as long as we prove the chain of connections from a Lie algebra down to its root space, the classification of root spaces gives us a classification of Lie algebras.

2 From Lie Groups to Lie Algebras

In this section we reduce the problem of classifying Lie groups to that of classifying complex simple Lie algebras. We assume many results beyond the scope of this paper, and skim over the details needed from topology, algebraic topology, and analysis to define the correspondences rigorously.

To any Lie group we can associate a Lie algebra, which is simpler to study because it is a vector space and we can apply our knowledge of linear algebra to it. Isomorphic Lie groups have isomorphic Lie algebras, but the converse is not true and two nonisomorphic Lie groups can have have isomorphic Lie algebras. However, the second-best thing is true:

1. Two *simply-connected* [1] Lie groups with isomorphic Lie algebras are isomorphic
2. Given a simply-connected Lie group, we can find all the connected Lie groups it is a universal cover of. This lets us find all other connected Lie groups with the same Lie algebra
3. We can understand non-connected Lie groups by understanding the connected component containing the identity

Before we can move forward, we need to review the concept of a covering space from algebraic topology.

Definition 2.1. *Let X and E be topological spaces and $p\colon E \to X$ be a continuous map. Then p is called a covering map, and E a cover of X if for every point $x \in X$ there exists a neighborhood U of x such that $p^{-1}(U)$ is the disjoint union $\bigsqcup U_i$ and each U_i is homeomorphic to U via p. A covering E is said to be the universal cover of X if E is simply-connected i.e. its fundamental group, $\pi_1(X)$, is trivial.*

[1] A simply-connected space is a path-connected space where every path can be continuously deformed (*homotoped*) into any other path. In other words, a simply connected space is a path-connected space with a trivial fundamental group.

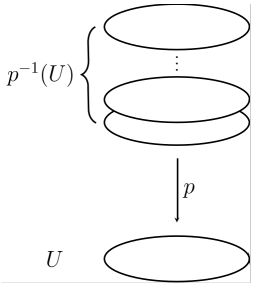


Figure 1: A covering map. Image credit Wikipedia

Intuitively, a covering space locally "looks like" the base space. An example is the map $p\colon \mathbb{R}^2 \to S^1 \times S^1$, $(x, y) \mapsto (x \mod 1, y \mod 1)$ that exhibits R^2 as a cover for the torus. Intuitively, we can think of the torus as the unit square $I \times I$ where if you go off one side, you wrap-around and come back on the other side. The cover is the space in which we do not wrap around, and the covering map sends us back to where we would have been had we wrapped around.

Every nice-enough[2] space has a universal cover. In particular, manifolds (and hence Lie groups) have a universal cover and we can put a unique group structure on the universal cover of a Lie group that makes the universal cover into a Lie group.

Theorem 2.2. *The Lie algebra of a Lie group G is isomorphic to the Lie algebra of its universal cover $\tilde{G}$. Moreover, if two connected Lie groups have isomorphic Lie algebras, their universal covers are isomorphic Lie groups.*

We shall not prove this theorem. Since simply-connected spaces are their own universal covers, we get that two simply-connected spaces with isomorphic Lie algebras are isomorphic. We now need to construct the non-simply connected Lie groups that have the same Lie algebra. To do this, we rely on the correspondence between covering spaces and fundamental groups in algebraic topology.

Definition 2.3. *Let $p\colon E \to X$ be a cover, then any automorphism f of E is called a deck transformation if $p \circ f = p$ i.e. it sends a point in $p^{-1}(x)$ to another point in $p^{-1}(x)$. It is easy to see that these form a group.*

Theorem 2.4 (Covering space - Fundamental group Galois Correspondence)**.** *Consider a space X with a universal cover $\tilde{X}$ and deck transformation group Γ. Then $X \cong X/\Gamma$ and $\Gamma \cong \pi_1(X)$. Moreover, for any other cover E of X, there exists a subgroup $\Gamma' \leq \Gamma$ such that $E \cong \tilde{X}/\Gamma'$. Two covering spaces are homeomorphic as covers if and only if their corresponding subgroups are conjugate.*

Suppose we have a simply-connected Lie group $\tilde{G}$ that is the universal cover of G via a map φ. Then $G \cong \tilde{G}/\ker(\varphi)$ (by the first isomorphism theorem), so by the correspondence theorem we know that $\pi_1(G) \cong \ker(\varphi)$. The fundamental group of a manifold is countable [3], so $\ker(\varphi)$ is countable and hence a discrete subgroup of $\tilde{G}$. This gives us a way of constructing G from $\tilde{G}$, if we can find all the discrete subgroups. This is not very difficult, by the following proposition

Proposition 2.5. *Any discrete normal subgroup of a path-connected topological group is central i.e. elements of the subgroup commute with any element of the whole group.*

Proof. Let $H \trianglelefteq G$ be the discrete normal subgroup, and suppose it isn't central. Then there exists some $h \in H$ such that $gh \neq hg \implies h \neq ghg^{-1}$. Since H is normal, $ghg^{-1} \in H$. Since g is a topological group, we can consider a path α from e (the identity) to g. Since H is normal, $\alpha(t)h\alpha(t)^{-1}$ is a pathlying entirely in H that goes from h to ghg^{-1}. This would give us uncountably many elements and a non-discrete topology, unless this is the trivial path, in which case $h = ghg^{-1}$. □

[2] connected, locally path connected, and semi-locally simply connected

[3] We define our manifolds to be second-countable. Then the countability of the fundamental group follows from this

Thus all we need to do is compute the center of the group (which is an abelian lie group, hence the direct product of a torus and euclidean space), and find its discrete subgroups. Lastly, we need to discuss the transition from connected to non-connected groups.

Proposition 2.6. *If G is a topological group and C is the connected component containing the identity, then C is a normal subgroup.*

Proof. Since G is a topological group, left- and right-multiplication by any element g is a continuous map. Its inverse is multiplication by g^{-1}, so it is a continuous map with a continuous inverse hence a homeomorphism. A homeomorphism maps open sets to open sets, closed sets to closed sets, and connected sets to connected sets, hence gC and Cg are open, closed, and connected i.e. they are a connected component. They contain g because $e \in C \implies ge \in gC$ and $eg \in Cg$. Thus $gC = Cg =$ the connected component containing the identity. We will now show that C is a normal subgroup of G.

- By definition $e \in C$
- If $g \in C$ then gC is the component containing g i.e. C, so $gC = C$ and hence for any $x \in C$, $gx \in C$.
- Inversion is a continuous bijection and its inverse (itself) is also a continuous bijection, so it sends C to a connected component. Since it sends the identity to itself, it must send C to C. Thus if $g \in C$, $g^{-1} \in C$.
- We showed above that for any $g \in G$, $gC = Cg =$ the connected component containing g. Thus $gCg^{-1} = C$.

□

Thus if G is a non-connected Lie group, we can quotient by the connected component of the identity to get G/C, a group structure on the connected component. Since manifolds have countably many components [4], this is a discrete countable group. Thus getting non-connected Lie groups from a connected one reduces to solving the extension problem for finite groups, and that is as far as we can go in this paper [5]

Thus the problem of classifying Lie groups boils down to the problem of classifying connected Lie groups, which further reduces to the problem of classifying simply-connected Lie groups, which have unique Lie algebras. There is one remaining hurdle to reducing the classification of Lie groups to that of Lie algebras: Are there Lie algebras that have no corresponding Lie group? Fortunately, the answer is no. To show this we need to assume the following theorem; for a proof, see here.

Theorem 2.7 (Ado's Theorem)**.** *Every Lie algebra $\mathfrak{g}$ is isomorphic to a lie subalgebra of $\mathfrak{gl}_n(\mathbb{R})$ for some n. In other words, every abstract lie algebra can be represented as a matrix lie algebra.*

This lets us prove the following theorem and establish a bijective equivalence between lie algebras and simply-connected lie groups:

Theorem 2.8 (Lie's Third Theorem)**.** *For any Lie algebra $\mathfrak{g}$, there exists a simply-connected Lie group G such that* $\mathrm{Lie}(G) = \mathfrak{g}$

Proof. Express $\mathfrak{g}$ as some subalgebra of $\mathfrak{gl}_n(\mathbb{R})$ and look at its image $e^{\mathfrak{g}}$ under the exponential map. Let G be the subgroup of $\mathrm{GL}_n(\mathbb{R})$ generated by this image. Because the exponential map is a diffeomorphism near the identity, $\mathrm{Lie}(G) = \mathfrak{g}$. G is not necessarily simply connected, so we consider its universal cover $\tilde{G}$. This gives us a simply-connected Lie group whose Lie algebra is $\mathfrak{g}$. □

[4] Manifolds are second-countable, which means that there is some countable set of open sets B_n such that every open set contains a B_i as a subset. Each connected component is itself as open set, so if each of them contained some B_i, the set $\{B_n\}$ would be uncountable, which contradicts second-countability

[5] but see this

3 Types of Lie Algebras

The problem, as outlined in the above section, is to now classify all Lie algebras. This problem turns out to be intractable (because of the difficult of classifying *solvable* Lie algebras), but we can still classify families of lie groups and algebras.

Our first concession is to only consider Lie alebras over the complex number field $\mathbb{C}$, because the algebraically-closed nature of $\mathbb{C}$ makes a lot of things much smoother. This ends up not affecting the overall classifcation that much: for every complex Lie algebra, there are real forms that complexify to it, so if we classify Lie algebras over the complex numbers, the only thing left to do is to understand this process of finding real forms.

Our second, more serious, concession is that we only classify a subtype of Lie algebras known as semisimple Lie algebras. The reason for this is twofold - semisimple Lie algebras have a very neat classification theory while general Lie algebras don't, and secondly that from a representation-theoretic point of view, semisimple Lie algebras are the only ones that matter. To explain this futher, we need to introduce some definitions.

Definition 3.1. *Let $\mathfrak{g}$ be a Lie algebra*

1. *For subalgebras $\mathfrak{a}, \mathfrak{b}$, define $[\mathfrak{a}, \mathfrak{b}]$ to be the subalgebra generated by $\{[a, b] \mid a \in \mathfrak{a}, b \in \mathfrak{b}\}$*
2. *Let $D^0\mathfrak{g} = \mathfrak{g}$, and $D^n\mathfrak{g} = [D^{n-1}\mathfrak{g}, D^{n-1}\mathfrak{g}$. This is known as the derived series of the Lie algebra.*
3. *Let $D_0\mathfrak{g} = \mathfrak{g}$, and $D_n\mathfrak{g} = [\mathfrak{g}, D_{n-1}\mathfrak{g}$. This is known as the lower central series of the Lie algebra.*
4. *Note that $D_1\mathfrak{g} = D^n\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$. We call this $D\mathfrak{g}$.*
5. *The center of a Lie algebra, $Z(\mathfrak{g})$, is the set of all elements a such that $[a, g] = 0$ for all $g \in \mathfrak{g}$*

These notions are analogous to similarly-named notions from group theory. Continuing the analogy, we bring in more terms that will help us break down Lie algebras.

Definition 3.2. *Let $\mathfrak{g}$ be a Lie algebra*

1. *Let $\mathfrak{a}$ be a subalgebra of $\mathfrak{g}$. Then $\mathfrak{a}$ is called an ideal of $\mathfrak{g}$ if $[\mathfrak{a}, \mathfrak{g}] = \mathfrak{a}$. In other words, if $a \in \mathfrak{a}$ and $g \in \mathfrak{g}$ then $[a, g] \in \mathfrak{a}$. This is similar to the notion of ideal encountered in ring theory. Note that $D\mathfrak{g}, D_n\mathfrak{g}$, and $Z(g)$ are all ideals.*
2. *Call a Lie algebra nilpotent if $D_n\mathfrak{g}$ eventually terminates in 0*
3. *Call a Lie algebra solvable if $D^n\mathfrak{g}$ eventually terminates in 0*
4. *Call a Lie algebra semisimple if it has no nonzero solvable ideals.*
5. *Call a Lie algebra simple if it has no nonzero ideals.*

The similarity in the names "simple" and "semisimple" is not a coincidence. As we shall see later, a semisimple Lie algebra is the direct sum of simple Lie algebras. Solvable and nilpotent Lie algebras are, in some sense, a generalization of abelian Lie algebras, whereas semisimple Lie algebras are the maximally non-abelian Lie algebras. We shall make these statements precise.

First we shall justify the narrowing of our focus to semisimple Lie algebras. It turns out that there is a way to decompose any Lie algebra into a semisimple component and a solvable component, though not necessarily as a direct sum. As we states previously, it is difficult to classify all solvable Lie algebras, but the classification of semisimple Lie algebras means that we can understand that component in any Lie algebra. Moreover (though we shall not prove this), irreducible representations of solvable Lie algebras are one-dimensional, and irreducible representations of general finite-dimensional Lie algebras depend essentially only on the semisimple component.

Theorem 3.3 (Levi Decomposition)**.** *There exists a "maximal solvable ideal" of* $\mathfrak{g}$*, called the radical and denoted* $\operatorname{Rad}(\mathfrak{g})$*. Then* $\mathfrak{g}/\operatorname{Rad}(\mathfrak{g})$ *is semisimple, and we get an exact sequence*

$$0 \longrightarrow \operatorname{Rad}(\mathfrak{g}) \longrightarrow \mathfrak{g} \longrightarrow \mathfrak{g}/\operatorname{Rad}(\mathfrak{g}) \longrightarrow 0$$

The proof relies on the existence of this maximal solvable ideal, so first we show that.

Theorem 3.4. *The sum of solvable ideals is again a solvable ideal. For every Lie algebra* $\mathfrak{g}$ *there exists a maximal solvable ideal* $\operatorname{Rad}(\mathfrak{g})$ *which is the sum of all the solvable ideals in it.*

Proof. We shall first need a result on quotient solvability. Suppose $\mathfrak{g}$ is a Lie algebra with a solvable subalgebra $\mathfrak{a}$ such that $\mathfrak{g}/\mathfrak{a}$ is again solvable. Let $\pi\colon \mathfrak{g} \to \mathfrak{g}/\mathfrak{a}$ be the projection operator. Then the solvability of $\mathfrak{g}/\mathfrak{a}$ implies that $\pi(D^n\mathfrak{g}) = 0$, or that $D^n\mathfrak{g} \in \mathfrak{a}$. Since $\mathfrak{a}$ is solvable, this means continuing the derived series onwards from here gives us a series that terminates in 0. Thus $\mathfrak{g}$ is solvable.

We are now ready to prove the main result. Suppose $\mathfrak{a}$ and $\mathfrak{b}$ are solvable ideals of $\mathfrak{g}$. Let $a \in \mathfrak{a}, b \in \mathfrak{b}, g \in \mathfrak{g}$. Then $[a+b, g] = [a.g] + [b, g] \in \mathfrak{a} + \mathfrak{b}$. Thus $\mathfrak{a} + \mathfrak{b}$ is an ideal. Now, the second isomorphism theorem from group theory gives us that $\mathfrak{a} + \mathfrak{b}/\mathfrak{a} \cong \mathfrak{b}/(\mathfrak{a} \cap \mathfrak{b})$. This latter quotient is solvable since both $\mathfrak{b}$ and $\mathfrak{a} \cap \mathfrak{b}$ are solvable. This means the the former quotient is too. Since $\mathfrak{a} + \mathfrak{b}/\mathfrak{a}$ is solvable and $\mathfrak{a}$ is solvable, quotient sovability implies that $\mathfrak{a} + \mathfrak{b}$ is also solvable.

Then we can just set $\operatorname{Rad}(\mathfrak{g})$ to be the sum of all solvable ideals in $\mathfrak{g}$. Thus it is maximal, and also contains every other solvable ideal. □

Proof of Levi decomposition. We need to show that $\mathfrak{g}/\operatorname{Rad}(\mathfrak{g})$ is semisimple, i.e., has no solvable ideals. Suppose it had a solvable ideal, then the lift of this ideal in $\mathfrak{g}$ is also solvable, by quotient solvability and the lattice isomorphism theorem. But then it would be included in $\operatorname{Rad}(\mathfrak{g})$ and go to zero in the quotient. □

Thus from now on, we shall focus only on semisimple Lie algebras.

4 Structure Theory of Lie Algebras

In this section we give a brief overview of the structure theory of Lie algebras, without a lot of proofs. The reason we care about the structure theory in the process of classification is that it is at this step that we can make the transition from Lie algebras to root systems.

To gain further insights into the structure of a Lie algebra, we consider consider the adjoint representation of ad_X of the Lie algebra. Recall that for $X \in \mathfrak{g}$, $\operatorname{ad}_X = [X, \cdot]$. Given the adjoint representation, we can define a useful bilinear form on $\mathfrak{g}$, the Killing form:

Definition 4.1. *Let* $B(X, Y) = \operatorname{Tr}(\operatorname{ad}_x \circ \operatorname{ad}_y)$*. In other words,* $B(X, Y)$ *is the sum of the eigenvalues of the linear operator* $[X, [Y, \cdot]]$*.*

Note that the Killing form also extends to the *dual* of $\mathfrak{g}^*$. If $\alpha, \beta \in \mathfrak{g}^*$, then we can set $B^*(\alpha, \beta) = B(t_\alpha, t_\beta)$ where t_α and t_β are the duals. Of course, this only works in the finite dimensional case, to which we are presently restricting.

The Killing form gives us a way to determine whether a Lie algebra is semisimple or solvable.

Theorem 4.2 (Cartan's Criterion)**.** *Let* $\mathfrak{g}$ *be a finite dimensional Lie algebra. Then*

1. $\mathfrak{g}$ *is solvable if and only if* $B(\mathfrak{g}, D\mathfrak{g}) = 0$
2. $\mathfrak{g}$ *is semisimple if and only if* B *is a non-degenerate form on* $\mathfrak{g}$

The Killing form is ultimately what will give us an inner-product structure on the root space. To define what roots are at all, we first need to consider Cartan subalgebras. Recall that for endomorphisms of vector spaces we can split them $\varphi = \varphi_s + \varphi_n$ into a semisimple (= diagonalizable) and a nilpotent component. This is known as the Jordan-Chevalley decomposition. It turns out that this plays well with representations in semisimple Lie algebras, particularly the adjoint representation. Call an element $x \in \mathfrak{g}$ semisimple (resp. nilpotent) if ad_x is semisimple (resp. nilpotent). Because of this decomposition, we know that there exists some element $x \in \mathfrak{g}$ which is semisimple (because if it we not, we could restrict to its semisimple component of the decomposition), and its span gives us a subalgebra of $\mathfrak{g}$ which consists of only semisimple elements.

Definition 4.3. *Call a subalgebra of $\mathfrak{g}$ toral if it consists only of semisimple elements. Let $\mathfrak{h}$ be the maximal toral subalgebra. Call $\mathfrak{h}$ the Cartan subalgebra of $\mathfrak{g}$.*

The most important property of the Cartan subalgebra is that it is abelian. Since it is abelian, it is *simultaneously diagonalizable* (commuting linear operators are simultaneously diagonalizable), and so we can decompose $\mathfrak{g}$ into eigenspaces of the action of $\mathfrak{h}$.

Definition 4.4. *Let $g_\alpha = \{v \in \mathfrak{g} \mid [h, v] = \alpha(h)v, h \in \mathfrak{g}\}$. In other words, g_α is the eigenspace with $\alpha \in \mathfrak{h}^*$ being the eigenvalue.*

The main theorem in the structure theory of semisimple Lie algebras is as follows:

Theorem 4.5. *Given a Cartan subalgebra $\mathfrak{h}$, it holds that $\mathfrak{g}_0 = \mathfrak{h}$ and there is a decomposition (as an $\mathfrak{h}$-module):*

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$$

where Φ is the set of all nonzero linear functionals α of $\mathfrak{h}$ such that $\mathfrak{g}_\alpha \neq \{0\}$. Moreover, for each $\alpha, \beta \in \Phi$,

1. *$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$, which is an equality if $\alpha + \beta \neq 0$.*
2. *$[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \oplus \mathfrak{g}_{-\alpha} \oplus \mathfrak{g}_\alpha \cong \mathfrak{sl}_2$ as a Lie algebra.*
3. *$\dim \mathfrak{g}_\alpha = 1$; in particular, $\dim \mathfrak{g} = \dim \mathfrak{h} + \#\Phi$.*
4. *$\mathfrak{g}_{2\alpha} = \{0\}$; in other words, $2\alpha \notin \Phi$.*
5. *With respect to the Killing form B, $\mathfrak{g}_\alpha, \mathfrak{g}_\beta$ are orthogonal to each other if $\alpha + \beta \neq 0$; the restriction of B to $\mathfrak{h}$ is nondegenerate.*

From this theorem we can derive properties of root systems that we shall see in the next section. These are derived by studying the representation theory of $\mathfrak{sl}_2(\mathbb{C})$.

5 Root Spaces

The benefit of doing all the calculations above and understanding the structure theory of Lie algebras is that we can get a set of roots of the Lie algebra. This is very useful, because it justifies a further narrowing of focus from Lie algebras to just their root systems, and lets us examine root systems removed from their origins as Lie algebras.

Here are the rules all root systems follow. We set them now to be the definition of an abstract root system:

Definition 5.1. *An* abstract root system *Φ in a Euclidean space V is a finite set of non-zero vectors (called* roots*) such that:*

1. *Φ spans V.*
2. *For any $\alpha \in \Phi$, the only scalar multiples of α in Φ are $\pm\alpha$.*
3. *Φ is closed under reflection: for $\alpha \in \Phi$, the reflection of β about the hyperplane perpendicular to α, $s_\alpha(\beta) = \beta - 2\frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle}\alpha$, is another root of Φ.*
4. *For $\alpha, \beta \in \Phi$, the quantity $\eta_{\beta,\alpha} = 2\frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle}$ is an integer.*

We can turn a concrete root system into an abstract one by letting V be the span of the roots $\alpha \in \mathfrak{h}^*$, and let $\langle \alpha, \beta \rangle = B(t_\alpha, t_\beta)$. The reverse process is a little more complicated but can also be done, so from any roo system we can produce a Lie algebra. Once we have abstracted away the Lie algebras, all we need to do is solve the geometric problem of classifying root systems.

First, we note that root systems are very rigid objects, with only certain allowable angles between roots.

Theorem 5.2. *The angle between two roots $\alpha \neq \pm\beta$ of a root system is one of $30°, 45°, 60°, 90°, 120°, 135°$ and $150°$.*

Proof. We know that $\eta_{\beta,\alpha} = 2\cos(\theta)\frac{||\alpha||}{||\beta||}$. To get rid of this factor, we just consider $\eta_{\beta,\alpha}\eta\alpha, \beta = 4\cos^2(\theta)$, which must be an integer, which gives us the restriction on angles that we needed. □

The main trick in the classification of root systems is the reduction of redundant information. We can easily remove some vectors from a root system, and recover them by *generating* them from the rest by means of the root system axioms. We shall try to understand the minimal structure necessary that will still let us recover the entire root system.

First, note that half the roots are redundant, because every root α comes paired with a $-\alpha$. We can thus take any hyperplane in E that does not contain any of the roots, and consider all the roots on one side of this hyperplace to be "positive roots" and the others to be "negative roots."

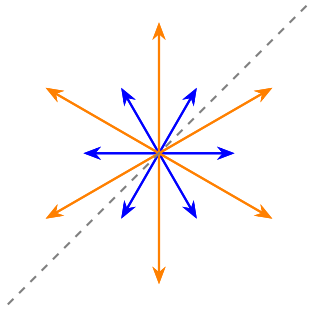


Figure 2: Dividing the roots into positive and negative roots

Restricting our focus to the positive roots, we can see that there must be a positive root that cannot be expressed as the sum of the other positive roots, since there must be at least one root that is closest to the hyperplane. This leads us to the following definition:

Definition 5.3. *Call a positive root simple if it cannot be expressed as the sum of two other positive roots*

It turns out that this is the right abstraction for reducing a root system into its constituent pieces. We shall now prove some properties of simple roots that will help us classify them.

Theorem 5.4. *If α and β are simple roots then the angle between them is not acute, i.e., $\langle \alpha, \beta \rangle \leq 0$*

Proof. Suppose $\langle \alpha, \beta \rangle > 0$, and WLOG suppose $\langle \alpha, \alpha \rangle \geq \langle \beta, \beta \rangle$. Now, since $\eta_{\alpha,\beta}\eta_{\beta,\alpha} < 4$, at least one of them is ± 1. Since that one will have the larger denominator, it has to be $\eta_{\beta,\alpha} = 2\frac{\langle \alpha,\beta \rangle}{\langle \alpha,\alpha \rangle}$. Since we assumed $\langle \alpha, \beta \rangle > 0$, this means $\eta_{\beta,\alpha} = 1$

Then, $s_\alpha(\beta) = \beta - \eta_{\beta,\alpha}\alpha = \beta - \alpha$ is a root, and so is $\alpha - \beta$. Exactly one of them is a positive root. If $\beta - \alpha$ is positive, this contradicts the simplicity of β, and if $\alpha - \beta$ is positive, it contradicts the simplicity of α. □

This immediately gives us some nice facts about the roots:

Theorem 5.5. *The simple roots are linearly independent.*

Proof. Suppose not, in which case there is a linear combination $\sum c_\alpha \alpha = 0$ ranging over the simple roots α. If all the c_α are nonpositive, then this cannot be zero, because all the α lie on one side of a hyperplane. Thus we can take all the negative coefficients to the other side and get

$$\sum_\alpha c_\alpha \alpha = \sum_\beta |c_\beta| \beta = v$$

Then consider

$$\langle v, v \rangle = \sum \sum |c_\alpha c_\beta| \langle \alpha, \beta \rangle$$

where every $\langle \alpha, \beta \rangle$ is nonpositive, so the overall sum is nonpositive, which cannot be since $\langle v, v \rangle$ must be greater than 0. □

Since the roots are lienarly independent and must also span Φ, there must be precisely n simple roots. The angle between any two simpe roots must be one of $90°, 120°, 135°$ and $150°$ by the non-acuteness criterion. We now turn to a powerful way of representing these simple root systems:

Definition 5.6. *A Dynkin diagram is graph where each node represents a simple root, and the lines between roots represents the angle between them. Arrows point from the larger root to the smaller.*

Angle between roots	Diagram edge	$\langle \alpha_1, \alpha_2 \rangle$
90°	• •	0
120°	•—•	$-\frac{1}{2}$
135°	•⇢•	$-\frac{\sqrt{2}}{2}$
150°	•⇉•	$-\frac{\sqrt{3}}{2}$

In the accompanying presentation, we prove the following result on the classification of Dynkin diagrams:

Theorem 5.7. *Every connected Dynkin diagram falls into one of four infinite families or one of five exceptions.*

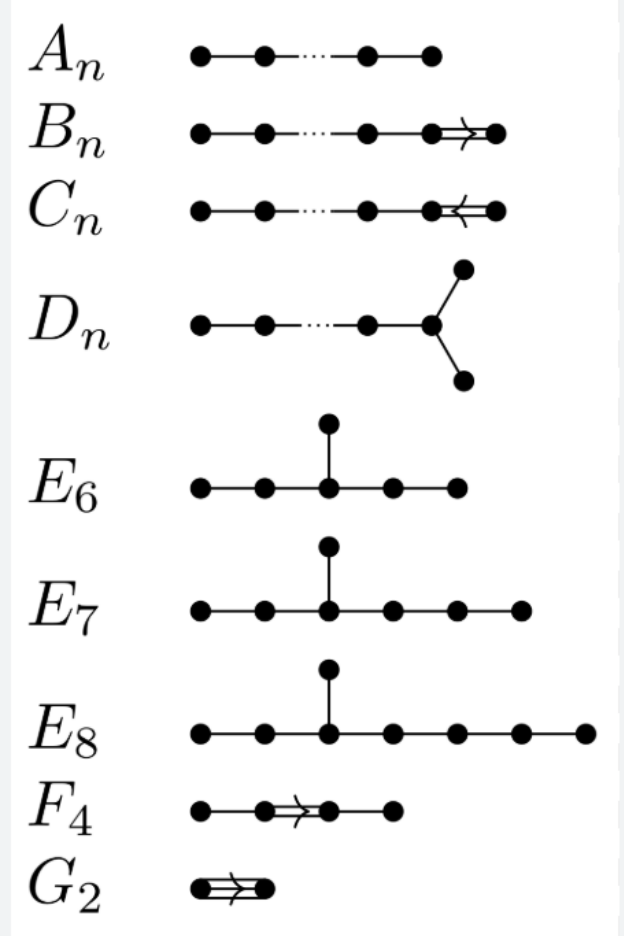


Figure 3: All connected Dynkin diagrams

6 Conclusion

Now that we have classified all the Dynkin diagrams, we can unwind the chain of reductions back to the Lie groups. Given any Dynkin diagram, we can associate it to a root system. From a root system we can get a semisimple complex Lie algebra. This is initially where our reductions stop, because we have only classified the semisimple Lie algebras. However, this still gives us much information.

For instace, while there are many real Lie algebras that complexify to a given semisimple Lie algebra, there is a unique "compact form," which leads to a unique simply connected compact Lie group. Thus, our classification of Lie algebras does give us the following classification of compact Lie groups,

Every compact connected Lie group has the form

$$G = \frac{S_1 \times \cdots \times S_m \times T}{Z}$$

where

- S_i are compact simple Lie groups
- Z is a discrete subgroup of the center of the semisimple portion
- T is a torus

and

$$\begin{array}{ccc} A_n \equiv & SU(n+1) & n \geq 1 \\ B_n \equiv & Spin(2n+1) & n \geq 2 \\ C_n \equiv & Sp(n) & n \geq 3 \\ D_n \equiv & Spin(2n+1) & n \geq 4 \\ G_2 & & \\ F_4 & & \\ E_6, E_7, E_8 & & \end{array}$$

where G_2, F_4, E_6, E_7, E_8 are exceptional Lie groups constructed directly so as to satisfy the root systems we saw.

7 References

1. *Representation Theory: A First Course*, Fulton and Harris.
2. *Lie Groups beyond an introduction*, Knapp.